Velocity AI: Insights Aggregator

Technical Architecture for Professional Agent Reporting & Strategic Growth

Overview

To achieve professional reporting, Velocity AI implements an **Insights Aggregator**. This microservice sits between the Instagram Graph API and MongoDB, transforming raw engagement metrics into actionable "learned" behaviors for the user.

Technical Execution Flow

1 Data Collection Layer

Pull raw metrics from the Instagram Graph API for every agent-generated post. These are persisted in a `metrics` collection in MongoDB Atlas.

- **Format Split:** Track `reach` and `impressions` specifically for Reels vs. Static vs. Carousels.
- **Temporal Analysis:** Log exact timestamps of `likes` and `comments` to identify high-velocity engagement windows.
- **Retention Metrics:** Track `saved` counts for Carousels to measure value-density.

2 The "Inference" Agent (Critic)

A specialized LangGraph node, the **Insight Generator**, runs weekly to synthesize data into human-readable brand intelligence.

- **Input:** 30-day window of raw JSON metrics from MongoDB.
- **Prompting:** *"Act as a Data Scientist. Find top 5 patterns where engagement is > avg. Identify specific times, days, or media formats."*
- **Goal:** Output structured insights that drive the UI "Learning" cards.

Implementation Roadmap

A. MongoDB Schema Design

Store isolated insights per brand to power the React dashboard experience:

```
{
  "brandId": "oreo_india_001",
  "agentInsights": [
    {
      "type": "format_performance",
      "text": "Reels consistently outperform static posts by 4x in reach",
      "confidence": 0.95
    },
    {
      "type": "timing",
      "text": "Posting between 7-9 PM drives 62% of your weekly engagement",
      "data_point": "19:00-21:00 UTC"
    }
  ],
  "currentStrategy": "70% Reels / 30% Carousels"
}
```

B. Frontend Strategy Shift (React)

When the agent identifies a performance gap (e.g., the 4x Reel advantage), it triggers two actions:

- **UI Notification:** Render the *"Agent Recommendation"* card to the user.
- **Backend Update:** Automatically patch the `post_frequency_config` in the database so next week's schedule prioritizes the winning format.

Visualizing "What Your Agent Learned"

In the MERN dashboard, map through the `agentInsights` array to create a dynamic feed:

- **Living Entity:** Use animated icons (brain/sparkles) to signify active learning.
- **Transparency:** Include a **Trust Score** showing the number of posts analyzed (e.g., *"Based on 42 recent posts"*).

70 LPA Pro-Tip: Don't just present data; present **Reasoning**. Use the AI to explain the "Why." *Example: "Reach is high on Wednesdays because your 'Food' niche trends for weekend-prep content."*